Soumiya Ayyappan_W9_D2

Easy qn

Step 1:







Step 2:

```
02:31 PM (1s)                                                    4

1 df=df.dropna(subset=["consultation_fee"])
2 df=df.withColumn("total_bill",col("consultation_fee")+col("medicine_fee"))
3 df=df.withColumn("review_flag",when(col("total_bill")>=1500,"HIGH").otherwise("NORMAL"))
4 display(df.count())
5 display(df)
```

See performance (2) — Checked for improvements

df: pyspark.sql.connect.dataframe.DataFrame = [bill_id: string, patient_id: string ... 7 more fields]

9

Table

	bill_id	patient_id	department	consultation_fee	medicine_fee	payment_status	bill_date
1	B001	P101	Cardiology	1200	450	PAID	2026-09-01
2	B002	P102	Orthopedics	900	300	PENDING	2026-09-01
3	B003	P103	General	500	120	PAID	2026-09-02
4	B005	P105	General	500	0	PAID	2026-09-03
5	B006	P106	Neurology	1600	550	PAID	2026-09-03
6	B007	P107	Orthopedics	900	200	PENDING	2026-09-04
7	B008	P108	Cardiology	1200	400	PAID	2026-09-04
8	B009	P109	General	500	150	PAID	2026-09-05
9	B010	P110	Neurology	1600	600	PENDING	2026-09-05

Step 3:

```
01:09 PM (8s)                                                    5

1 df.write.format("delta").mode("overwrite").saveAsTable("uc_practice_health.billing_ops.patient_billing_silver")
```

See performance (1) — Checked for improvements

```
01:17 PM (4s)                                                    6                                    Python

1 display(spark.sql("select count(*) as total_count from uc_practice_health.billing_ops.patient_billing_silver"))
2 display(spark.sql("select count(*) as department_count,sum(total_bill) from uc_practice_health.billing_ops.patient_billing_silver group by
  department order by department"))
```

See performance (2) — Checked for improvements

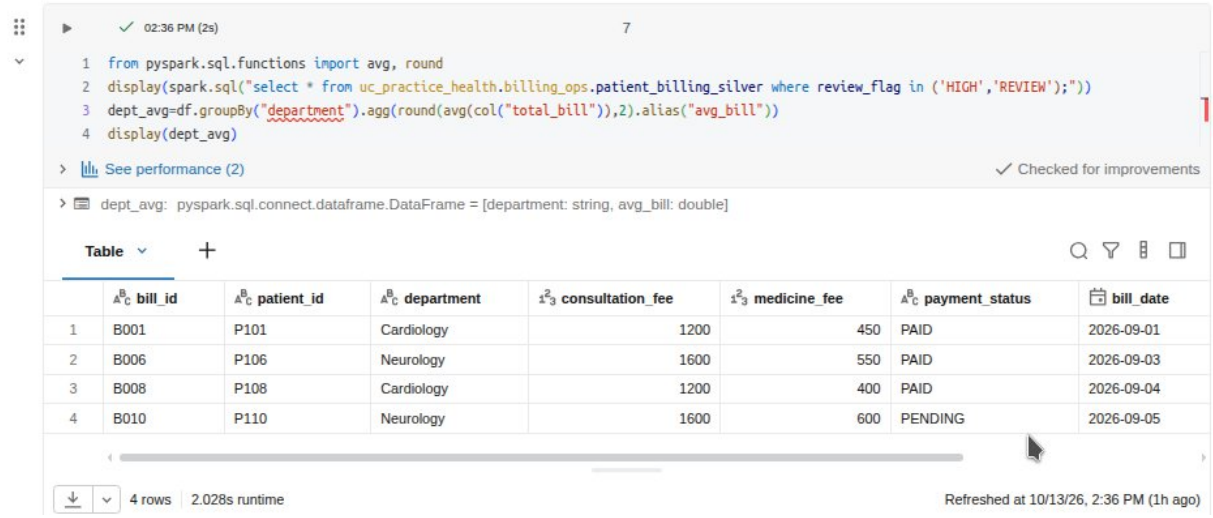
Table

	total_count
1	9

Table

	department_count	sum(total_bill)
1	2	3250
2	3	1770
3	2	4350
4	2	2300

Step 4:



02:36 PM (2s) 7

```
from pyspark.sql.functions import avg, round
display(spark.sql("select * from uc_practice_health.billing_ops.patient_billing_silver where review_flag in ('HIGH','REVIEW');"))
dept_avg=df.groupBy("department").agg(round(avg(col("total_bill")),2).alias("avg_bill"))
display(dept_avg)
```

See performance (2) Checked for improvements

dept_avg: pyspark.sql.connect.dataframe.DataFrame = [department: string, avg_bill: double]

Table +

	bill_id	patient_id	department	consultation_fee	medicine_fee	payment_status	bill_date
1	B001	P101	Cardiology	1200	450	PAID	2026-09-01
2	B006	P106	Neurology	1600	550	PAID	2026-09-03
3	B008	P108	Cardiology	1200	400	PAID	2026-09-04
4	B010	P110	Neurology	1600	600	PENDING	2026-09-05

4 rows | 2.028s runtime Refreshed at 10/13/26, 2:36 PM (1h ago)

Table +

	department	avg_bill
1	Orthopedics	1150
2	Cardiology	1625
3	General	590
4	Neurology	2175